

Advantages of the Markowitz Model.

- fairly systematic.
- effectively helps to reduce investment risks while potentially maximizing portfolio returns due to the EF.
- assumes diversification of uncorrelated assets.
- investors can construct a balanced portfolio that aligns with their risk tolerance and financial goals, promoting efficient and strategic investment decisions.

Limitation of the Markowitz Model.

- depends on historical data, which may not reliably predict future market trends.
- underlying assumptions are predicted on normally functioning markets.
- assets that do not follow a normal distribution will not work.
- Estimating a massive covariance matrix is computationally expensive.
- The Markowitz model does not provide guidelines for the expected return on assets.

a Single Index Model (= one-factor model = a market model)

$$r_i = E(r_i) + \beta_i m + \epsilon_i$$

r_i : return of asset i . (var: σ_i^2)

β_i : coef. of asset sensitivity to macro condition.

m : factor of macro condition (e.g., fluctuation of interest rate, $\begin{matrix} \text{mean: } 0 \\ \text{var: } \sigma_m^2 \end{matrix}$)

ϵ_i : error (mean 0, var $\sigma^2(\epsilon_i)$)

Suppose that $\text{Cov}(r_i, m) = 0$, $\text{Cov}(\epsilon_i, \epsilon_j) = 0$ ($i \neq j$).

$$\sigma_i^2 = \sigma_m^2 + \sigma^2(\epsilon_i)$$

$$\text{Cov}(r_i, r_j) = \text{Cov}(\beta_i m + \epsilon_i, \beta_j m + \epsilon_j) = \beta_i \beta_j \sigma_m^2$$

When considering individual stocks as assets, a single-factor model can be applied using a stock price index such as KOSPI as a macroeconomic variable.

$$R_i(t) = \alpha_i + \beta_i R_M(t) + \epsilon_i(t), \quad R_i = r_i - r_f, \quad R_M = r_M - r_f$$

α_i : Expected excess return of stock i .

β_i : Sensitivity of stock i for index.

ϵ_i : Error term.

Relation of expected return with beta:

$$E(R_i) = \alpha_i + \beta_i E(R_M)$$

α_i : Nonmarket Premium

β_i : Systematic Risk Premium

The excess return of a portfolio invested in all assets with equal weight is

$$R_p = \alpha_p + \beta_p R_M + \epsilon_p$$

$$R_p = \frac{1}{n} \sum_i R_i = \frac{1}{n} \sum_i (\alpha_i + \beta_i R_M + \epsilon_i) = \alpha_p + \beta_p R_M + \epsilon_p$$

$$\alpha_p = \frac{1}{n} \sum_i \alpha_i$$

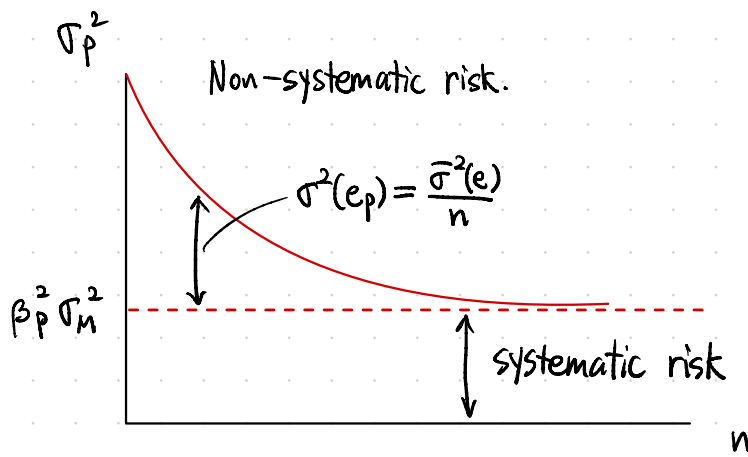
$$\beta_p = \frac{1}{n} \sum_i \beta_i$$

$$\epsilon_p = \frac{1}{n} \sum_i \epsilon_i$$

$$\sigma_p^2 = \beta_p^2 \sigma_M^2 + \sigma^2(\epsilon_p)$$

Systematic Risk : $\beta_p^2 \sigma_M^2$

Non-systematic Risk : $\frac{\sigma^2(\epsilon)}{n}$



Security Characteristic Line (SCL, 증권특성선)

SCL of a company A:

$$R_A(t) = \alpha_A + \beta_A R_M(t) + e_A(t)$$

$R_A(t)$: time series data on excess return of stock A

$R_M(t)$: " of Index (e.g., KOSPI)

→ Estimate α_A and β_A by regression analysis or etc.